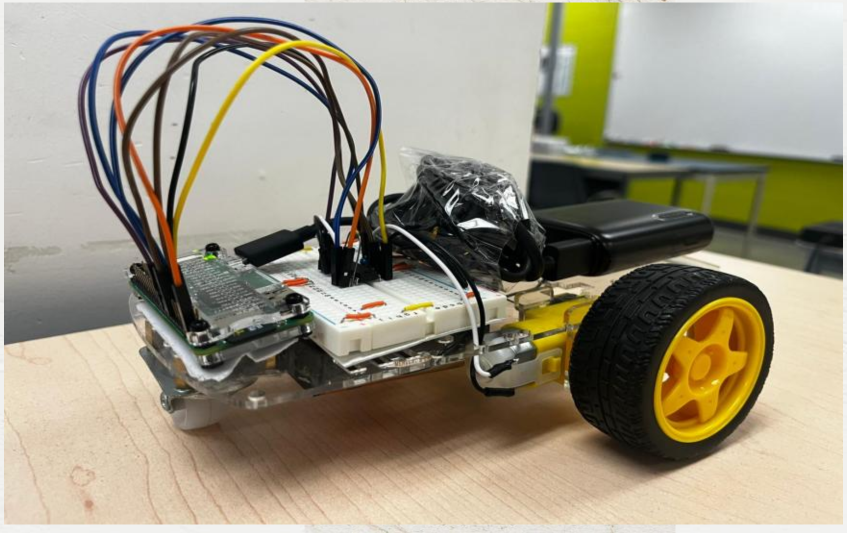
RoboRide



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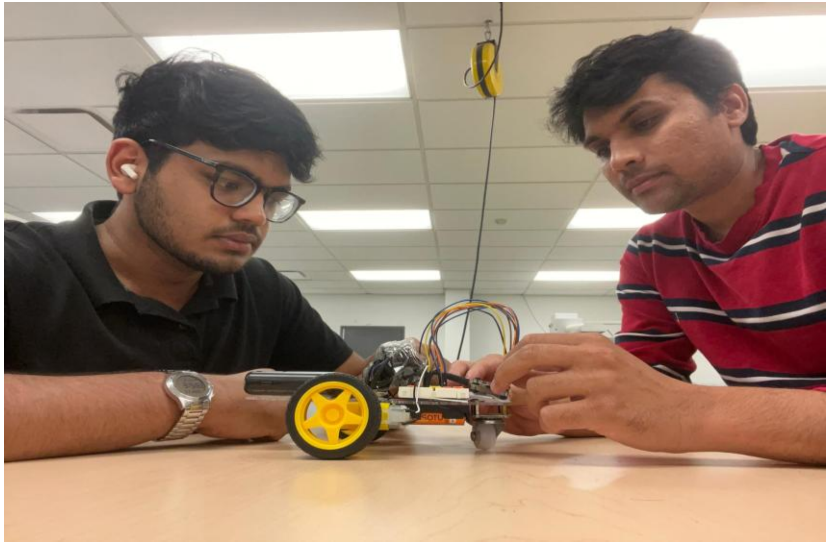
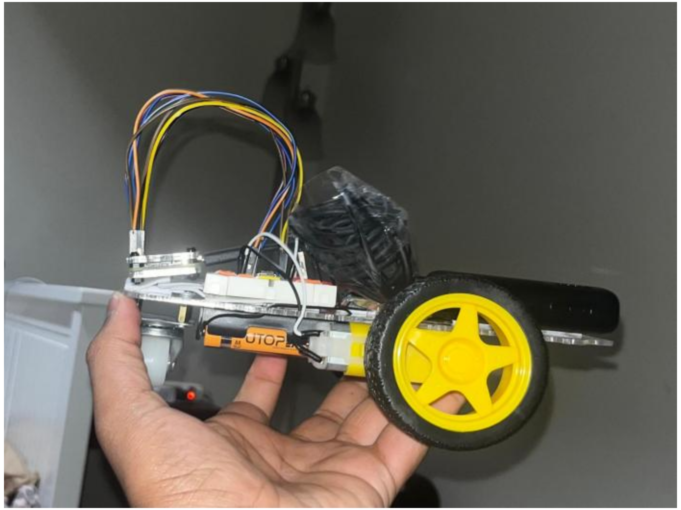
Introduction

- According to today's market trends, there is a strong need for compact, intelligent transport solutions and for tools that inspire the next generation of innovators. RoboRide is our answer to both.
- As a small autonomous transport vehicle, RoboRide can be used in environments like hospitals, factories, and campuses for safe and efficient delivery. At the same time, it can be introduced to students as an educational platform — a hands-on way to learn robotics, coding, and creativity.
- **RoboRide is not just a product. It's a bridge between solving real-world problems today and empowering the innovators of tomorrow.**



Project Overview

- **RoboRide** is a compact autonomous transport robot built to handle small-scale delivery and inspire creativity.
- **Smart Mobility:** Equipped with sensors and intelligent navigation, RoboRide can move goods safely across hospitals, warehouses, campuses, or offices.
- **Dual Purpose:** Beyond industry, RoboRide doubles as an educational robotics platform, giving students hands-on experience with coding, design, and problem-solving.
- **Affordable & Scalable:** Designed for low cost, adaptability, and easy customization, making it accessible for both businesses and schools.
- **Vision:** To redefine short-distance transport while sparking the next generation of innovators.



Project Purpose

- **Efficiency in Industry** – RoboRide is designed to provide a **safe, reliable, and cost-effective solution for short-distance transport** of goods within hospitals, warehouses, and campus environments. By automating repetitive delivery tasks, it reduces human workload, minimizes errors, and increases overall operational efficiency.

- **Creativity in Education** – Beyond industrial applications, RoboRide serves as an **educational platform that sparks innovation and curiosity among students**. It allows learners to explore robotics, programming, and problem-solving through hands-on experience.

Project Goals

- **Develop a functional autonomous prototype** with intelligent navigation and modular design.
- **Demonstrate real-world applications** through pilot use cases in logistics, healthcare, and academic settings.
- **Ensure affordability and scalability** so RoboRide can be produced at low cost and adapted for different industries.
- **Engage students and educators** by introducing RoboRide as a learning tool in robotics, coding, and engineering.
- **Lay the foundation for commercialization** with a clear path toward market entry and investor opportunities.

Technical Components

- **Controller:** Raspberry Pi for processing and motor control.
- **Motor Driver:** L293D used to drive two DC motors.
- **Power Supply:** 5000 mAh power bank (for Pi) + 4 AA batteries (for motors).
- **Chassis:** Compact car frame with two DC motors and tyres.
- **Connections:** Breadboard and jumper wires for easy circuit setup.
- **Functionality:**
- Robot moves forward, backward, left, and right using PWM control.
- Commands sent via terminal for precise motion.
- Efficient, stable, and expandable for automation or education.

Duty Cycles

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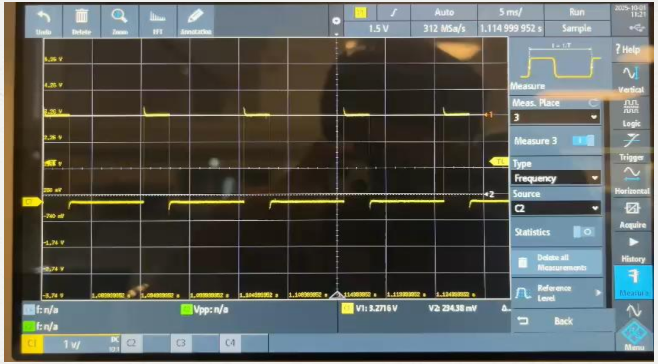

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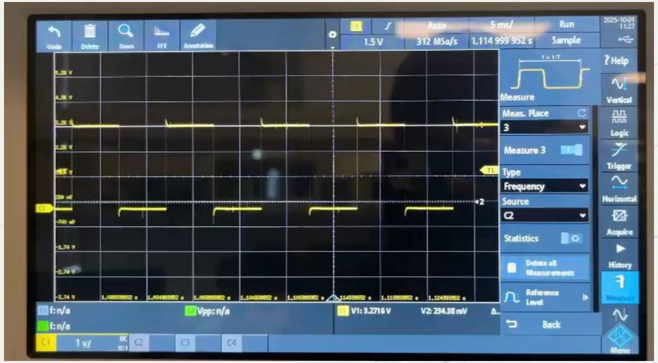

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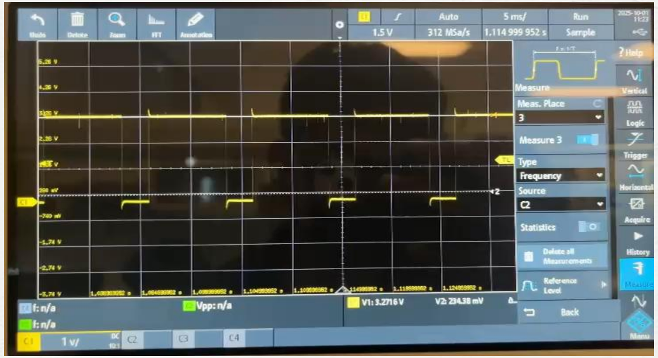

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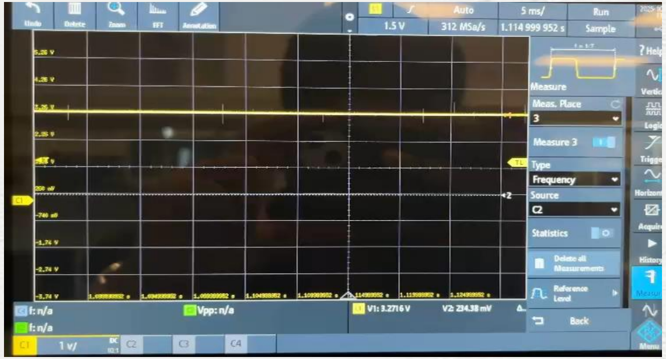
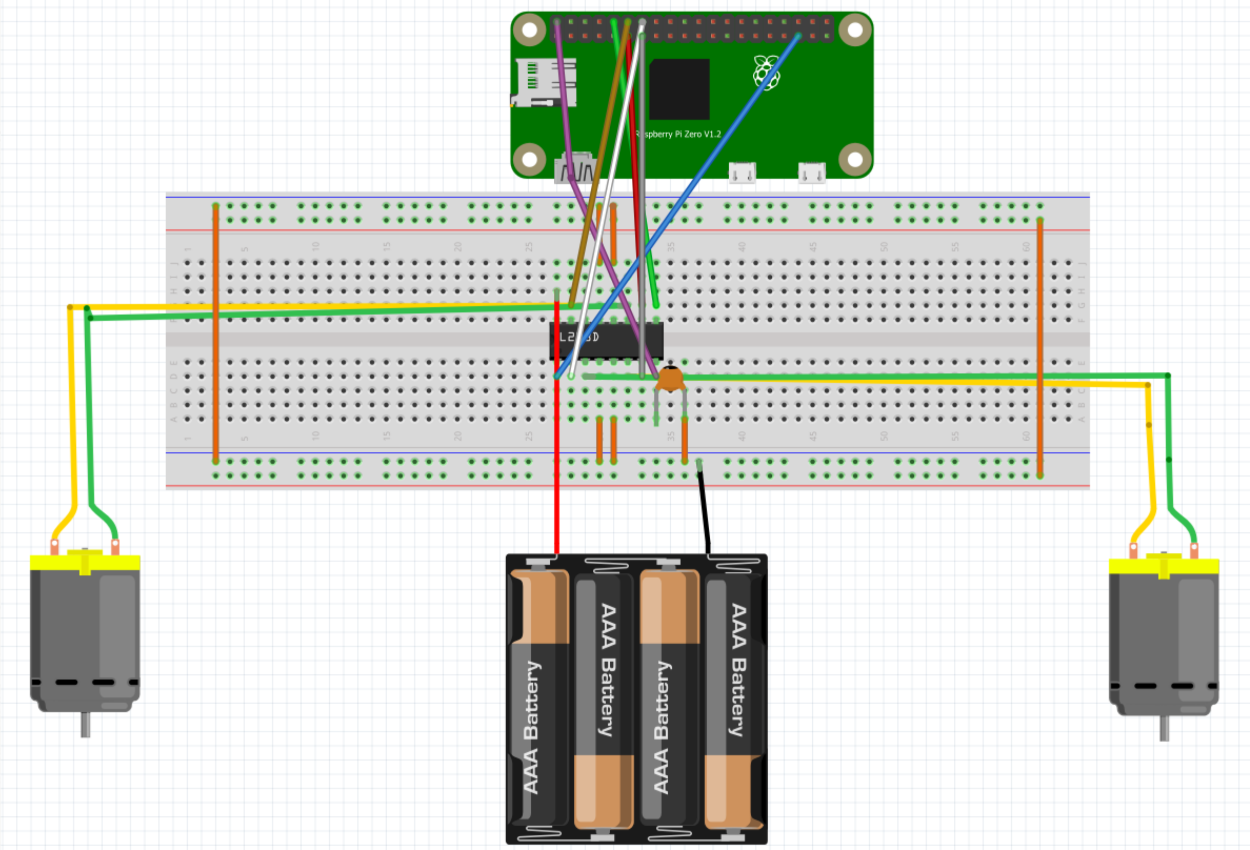


Bill of materials

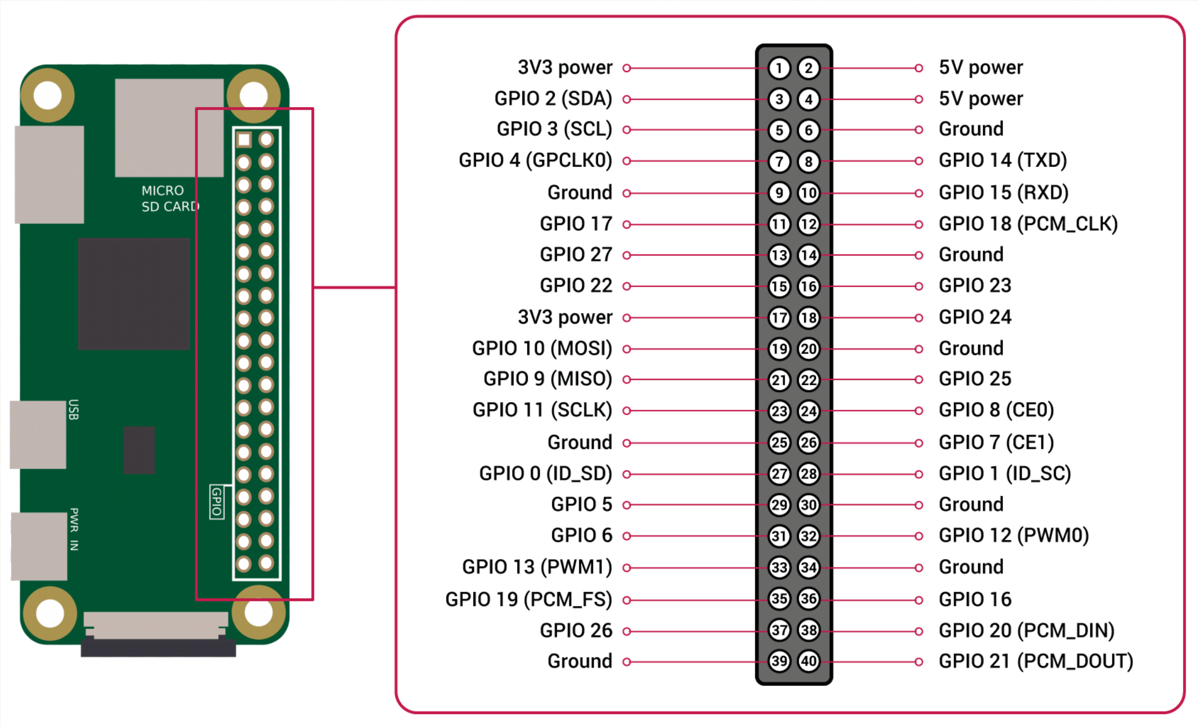
#	Component	Description	Qty	Unit Price (CAD)	Subtotal (CAD)
1	**Raspberry Pi 4 (4 GB)**	Main controller for GPIO and PWM control	1	80.00	80.00
2	**L293D Motor Driver IC**	Dual H-Bridge for controlling two DC motors	1	5.00	5.00
3	**DC Motors (3–6 V)**	Drives left and right wheels	2	6.00	12.00
4	**Robot Chassis with Wheels**	Base platform and mounting for motors	1	15.00	15.00
5	**Breadboard + Jumper Wires Set**	For circuit prototyping and GPIO connections	1 set	10.00	10.00
6	**6 V Battery Pack & Holder**	Separate power source for motors	1	8.00	8.00
7	**Misc. Hardware (screws, mounts, clips)**	Assembly and support components	1 set	5.00	5.00

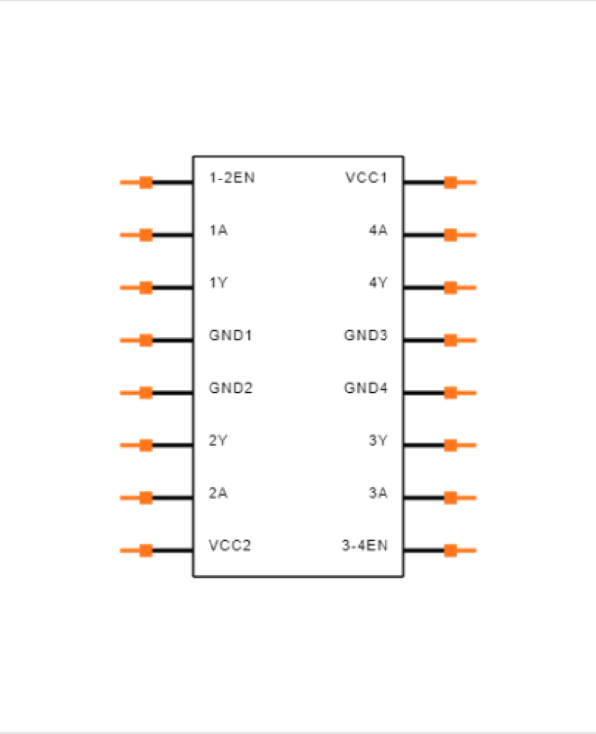
Total Price = $135 only

Wiring Diagram



IC & Pi Wiring Help





Market Oppurtunity

Applications:

- **Healthcare:** Delivering medication or samples safely within hospitals.
- **Logistics & Manufacturing:** Transporting components across factory floors.
- **Education:** Used in schools/universities as a robotics teaching tool.

Unique Selling Points (USPs):

- Dual-purpose (industrial + educational).
- Modular, affordable, and customizable.
- High adaptability to multiple industries

Investment Proposition

- **Dual Market Reach:** Serves both industrial automation and education sectors.
- **Low-Cost & Scalable:** Affordable to produce, easy to adapt for various uses.
- **High ROI Potential:** Low production cost (~$135) with strong profit margins.
- **Growth Opportunity:** Rising demand for smart delivery robots and STEM tools.
- **Call to Action:** Invest in RoboRide — powering the future of smart mobility and learning.

Challenges Faced and Solutions Implemented

- Faced difficulty connecting the Raspberry Pi and setting up GPIO pins correctly.
- Encountered motor polarity issues, causing the wheels to spin in opposite or incorrect directions.
- Wiring was congested on the breadboard, making troubleshooting and circuit organization difficult.
- Required multiple testing and rewiring sessions to stabilize connections and ensure consistent performance.
- Improved layout and cable management for a cleaner, more reliable build.

Conclusions

- RoboRide represents a powerful fusion of innovation, practicality, and education. It addresses real-world challenges in logistics by providing a safe, efficient, and cost-effective transport solution, while simultaneously serving as an educational platform that inspires creativity and hands-on learning in STEM fields. With its dual-purpose design, scalable technology, and strong market potential, RoboRide stands as both a viable business opportunity and a catalyst for future innovation.
- **Invest in RoboRide — driving smarter mobility and empowering the innovators of tomorrow.**

Future Improvements

Add AI-based navigation and obstacle detection

Implement IoT monitoring and mobile control interface

Improve battery efficiency and wireless charging

Support heavier payloads with modular chassis design

Thank you

“RoboRide isn’t just a robot — it’s a platform for innovation, learning, and smarter mobility.
Your investment powers the next step toward an intelligent and inspired future.”

RoboRide

Bridge between solving problems and empowering innovators.